

SAPIENZA
UNIVERSITÀ DI ROMA

Space and astronautical engineering

Homework 2

Lyapunov orbits

Professor:
Christian Circi

Student:
Girolamo Castronovo

Academic Year 2025/2026

Contents

1 Introduction **2**

2 Invariant Manifolds: Theory and Computation **3**

2.1 Numerical Approximation 4

3 The Poincaré Sections **6**

3.1 Manifolds on the Section 6

3.2 Sections Selected for the Analysis 6

3.2.1 Vertical Section (Σ_{vert}) 7

3.2.2 Horizontal Section (Σ_{horz}) 7

4 Trajectory Design and Classification **8**

4.1 Transit Trajectory: $\{X, [M], E\}$ 8

4.2 Transit Trajectory: $\{E, [M], X\}$ 11

4.3 Non-Transit Trajectory: $\{X, [M], X\}$ 13

4.4 Non-Transit Trajectory: $\{E, [M], E\}$ 16

1 Introduction

This report presents a comprehensive analysis of the stability properties and the associated phase space structures of Lyapunov orbits within the framework of the Earth-Moon Planar Circular Restricted Three-Body Problem (PCR3BP). Building upon the computation of periodic solutions around the collinear libration points L_1 and L_2 performed in the previous homework, this work focuses on the generation of Invariant Manifolds and their application to trajectory design with prescribed itineraries.

The stability of the periodic orbits is first evaluated through the analysis of the Monodromy matrix $M(T, 0)$. The eigenspectrum of this matrix reveals the hyperbolic nature of the equilibrium, characterized by the existence of stable and unstable subspaces. By introducing small perturbations along the associated eigenvectors, it is possible to approximate and propagate the global Stable (W^S) and Unstable (W^U) manifolds. These structures act as dynamical separatrices in the phase space, governing the natural transport of mass within the system and enabling low-energy ballistic transfers.

The core objective of this study is to exploit these manifolds to construct trajectories with specific topological behaviors. By selecting a pair of Lyapunov orbits around L_1 and L_2 characterized by the same Jacobi constant ($C \in [3.1370, 3.1493]$), the intersection of their manifolds is analyzed using Poincaré maps. This approach allows for the identification of initial conditions corresponding to four distinct itineraries through the Earth (E), Moon (M), and Exterior (X) realms:

- Transit trajectories: $\{X, [M], E\}$ and $\{E, [M], X\}$;
- Non-transit trajectories: $\{X, [M], X\}$ and $\{E, [M], E\}$.

The notation $[M]$ indicates that the initial conditions are defined within the Moon realm, specifically on the Poincaré section, before propagating forward and backward to verify the complete itinerary.

2 Invariant Manifolds: Theory and Computation

The stable (E^S) and unstable (E^U) subspaces, computed via the eigenvectors of the Monodromy matrix, represent the linear approximation of the local dynamics. In the full nonlinear model of the CR3BP, these subspaces extend into global surfaces called Invariant Manifolds. A manifold implies that any trajectory starting on this surface will naturally remain confined to it for its entire evolution.

For a periodic orbit Γ, we define two distinct types of manifolds based on the asymptotic behavior of the flow:

- **Stable Manifold (W^S):** It is the set of all trajectories that asymptotically approach the periodic orbit Γ as time tends to infinity ($t \to +\infty$). These trajectories naturally "flow" towards the orbit.

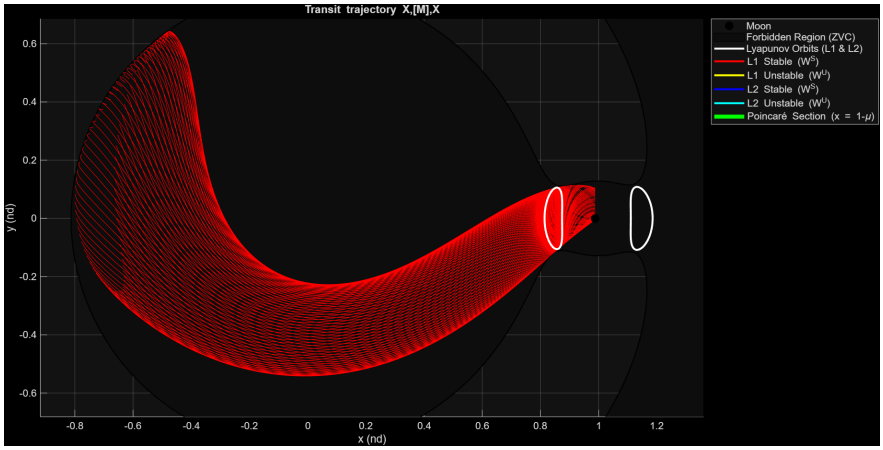


Figure 1: L1 Stable

- **Unstable Manifold (W^U):** It is the set of all trajectories that asymptotically depart from the periodic orbit Γ as time tends to minus infinity ($t \to -\infty$). In forward time, these trajectories drift away from the orbit.

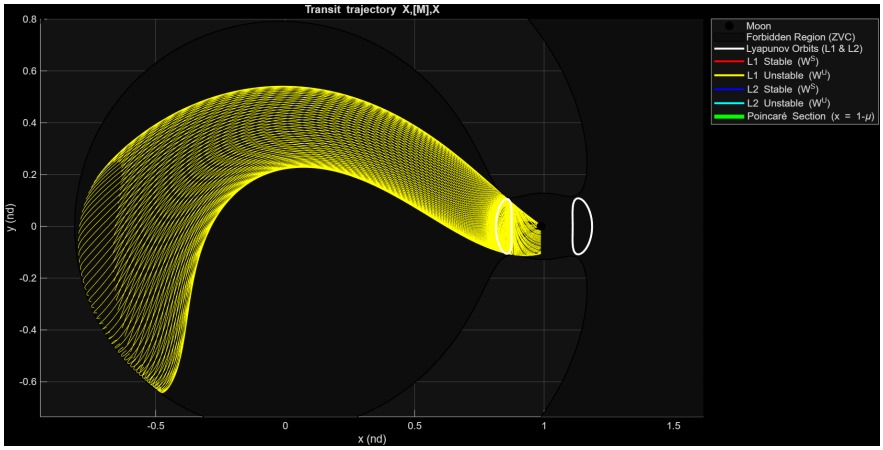


Figure 2: L1 Unstable

In the spatial projection (2D), these manifolds resemble four-dimensional tubes that emanate from the periodic orbit. They act as separatrices in the phase space, delimiting regions with different dynamical behaviors and governing the natural transport of matter through the Lagrange points.

2.1 Numerical Approximation

Since no analytical solution exists for the manifolds in the CR3BP, they must be computed numerically. The computation exploits the local linear dynamics to generate an initial guess, which is then propagated using the full nonlinear equations of motion.

The procedure adopted in this work consists of the following steps:

1. **Discretization:** The periodic orbit Γ is discretized into a set of N fixed points $X_\Gamma(t_i)$. (In this case N=100)

2. **Eigenstructure Analysis:** For each point on the orbit, the Monodromy matrix $M(t_i)$ is evaluated to extract the stable (V^S) and unstable (V^U) eigenvectors associated with the hyperbolic nature of the orbit.

3. **Perturbation:** To initialize a trajectory on the manifold, the state is perturbed slightly away from the equilibrium orbit along the direction of the eigenvector. The initial state X_0 is given by:

$$X_0 = X_\Gamma(t_i) \pm \epsilon \frac{V^{S,U}}{||V^{S,U}_{pos}||} \tag{1}$$

where ϵ is a small displacement (to ensure that the state remains within the linear region) and $||V^{S,U}_{pos}||$ represents the norm of the position components of the eigenvector.

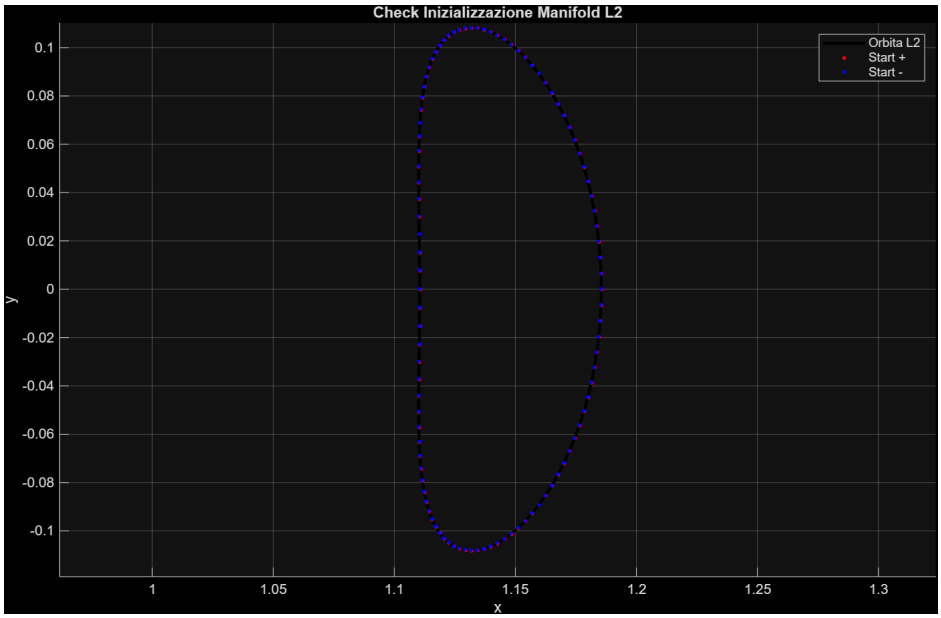


Figure 3: L2 orbit initial state +-

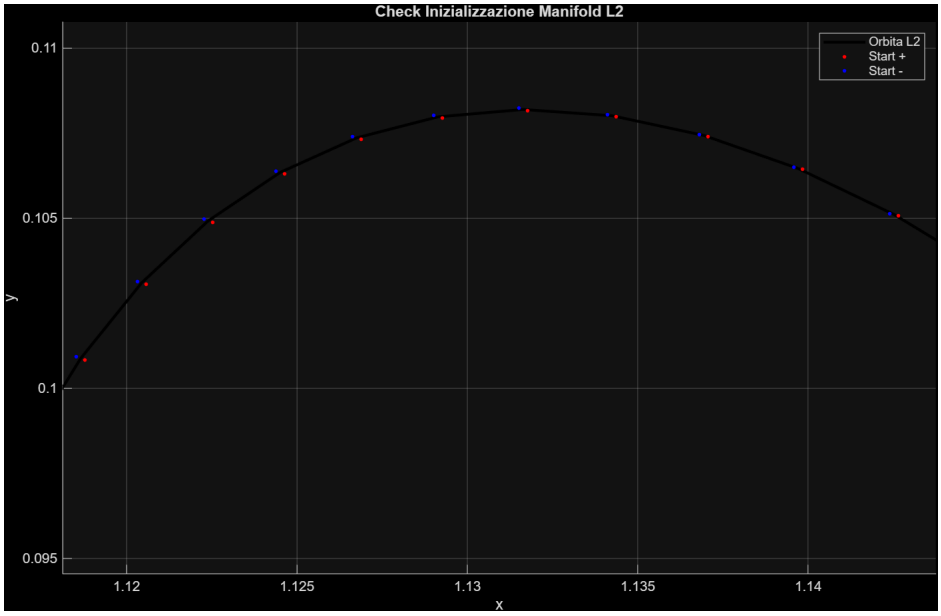


Figure 4: L2 initial states +- zoom

4. **Propagation:** The direction of integration depends on the type of manifold:

   - The **Unstable Manifold (W^U)** is generated by propagating the perturbed state forward in time $[0; T]$.
   - The **Stable Manifold (W^S)** is generated by propagating the perturbed state backward in time $[0; -T]$.

By repeating this procedure for both the positive $(+\epsilon)$ and negative $(-\epsilon)$ directions along the eigenvectors (corresponding to the two "branches" of the manifold), the complete tubular structure governing the flow entering and exiting the libration point region is reconstructed.

3 The Poincaré Sections

In the context of the Circular Restricted Three-Body Problem (CR3BP), the state of a spacecraft is defined by a four-dimensional vector composed of: position (x, y) and velocity (u, v) components:

$$\mathbf{X} = [x, y, u, v] \tag{2}$$

Visualizing the dynamical evolution of a trajectory within a four-dimensional phase space presents significant challenges. To address this complexity and facilitate the analysis of the dynamics, the method of Poincaré Sections is employed, effectively reducing the dimensionality of the problem from 4D to 2D.
The reduction of the state space dimensions is achieved by exploiting two fundamental physical constraints inherent to the system:

1. **Energy Conservation:** The Jacobi Constant (C) is fixed for a given trajectory. Consequently, the four state variables are not independent; knowing three variables allows for the determination of the fourth via the energy integral:

$$u^2 + v^2 = 2U(x, y) - C \tag{3}$$

   This constraint restricts the dynamics to a three-dimensional energy surface.

2. **The Poincaré section:** A specific geometric plane, denoted as Poincaré section (Σ_P), is defined within the configuration space (e.g., by fixing the coordinate $x = x_\Sigma$). The state of the system is recorded exclusively when the trajectory intersects this plane.

By fixing the Jacobi constant and constraining one spatial coordinate, the system dynamics can be fully represented by plotting the remaining two variables (typically position and velocity components) on a two-dimensional map.

3.1 Manifolds on the Section

The Invariant Manifolds (W^S and W^U) describe tubular structures in the phase space. The intersection of these three-dimensional tubes with a two-dimensional plane results in a closed curve. Studying these curves is fundamental for the development of transit and non-transit orbits.

3.2 Sections Selected for the Analysis

To identify the specific itineraries prescribed for this study, two distinct Poincaré sections were selected:

3.2.1 Vertical Section (Σ_{vert})

A vertical plane defined by $x = 1 - \mu$, located in the center of the Moon. This section is optimal for intercepting horizontal flows and is, therefore, utilized to identify Transit Orbits (such as $\{X, M, E\}$) that pass by the neck regions.

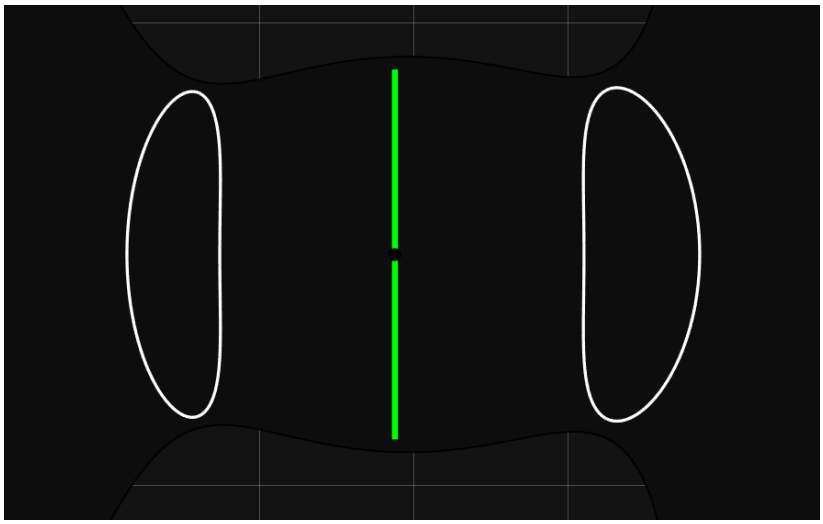


Figure 5: Vertical Poincaré Section

3.2.2 Horizontal Section (Σ_{horz})

A horizontal plane defined by $y = 0$. This section is useful for capturing trajectories. It is employed to identify NonTransit Orbits (such as $\{X, M, X\}$) that perform revolutions around the Moon without exiting the region immediately.

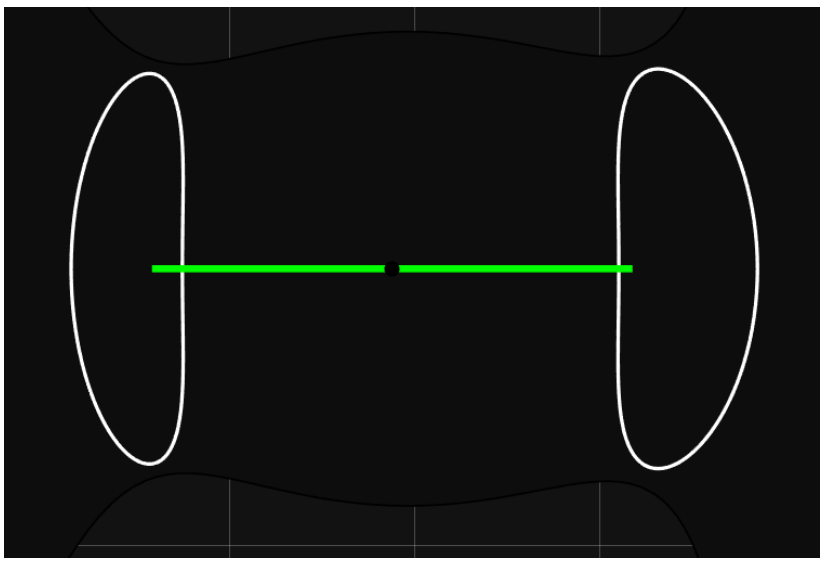


Figure 6: Horizontal Poincaré Section

4 Trajectory Design and Classification

The computation of specific itineraries relies on the division of the space into distinct "realms" separated by the neck regions around the Lagrange points. The Earth realm is denoted as E, the Moon realm as M (located between L_1 and L_2), and the Exterior realm as X. By computing the invariant manifolds as transport tubes, it's possible to evaluate transfers between these regions.
The method followed for the design of these trajectories involves the identification of intersections of the manifolds on the Poincaré sections specified earlier. If a state on the section belongs simultaneously to the unstable manifold of the departure orbit and the stable manifold of the arrival orbit, a continuous ballistic trajectory is possible between the two dynamical states.

4.1 Transit Trajectory: $\{X, [M], E\}$

This itinerary represents an inbound capture trajectory, where the spacecraft is coming from the Exterior realm, transits through the Moon region, and enters the Earth realm. To achieve this transfer, the trajectory must utilize the L_2 gateway to enter M and the L_1 gateway to exit into E.
Evaluating the manifolds necessary for this transfer is important to understand the boundaries of the latter, as long as the spacecraft is located inside the manifolds, its orbit will follow the natural flow.

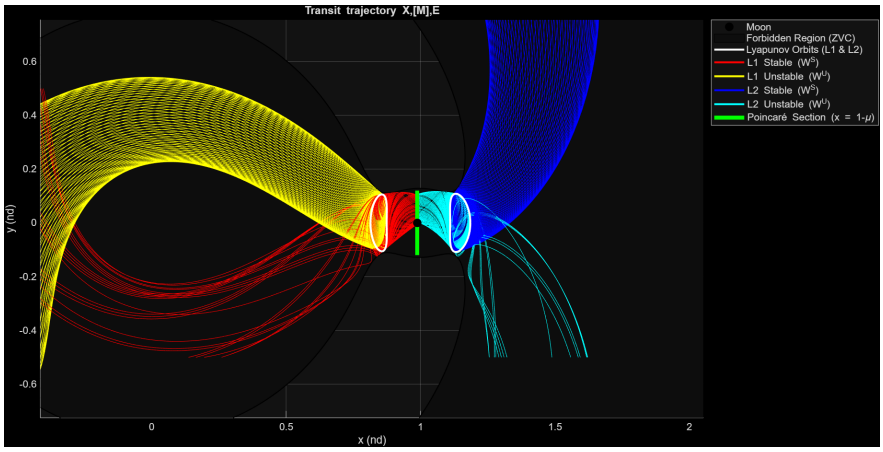


Figure 7: Transit X,M,E

In order the spacecraft will move inside the manifold of L2 Unstable, approaching the L2 point, then it will follow the path of L2 Stable inside the M realm, towards the vertical Poincaré section, here a calculation is performed. To understand the connection between the orbits around L1 and L2 it's necessary to analyze the Poincaré maps.

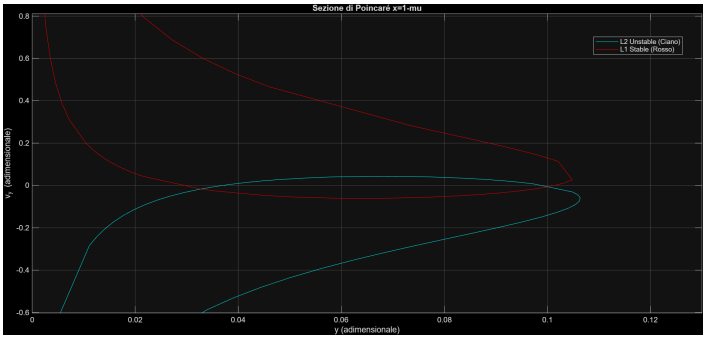


Figure 8: Poincaré Map (y vs v)

This plot shows clearly the intersection between L2 unstable manifold and the L1 stable, each value of (y,v) inside the common area is a connection state, the value of C is known and the value of x is fixed, so from choosing a point inside this region is possible to have a full state.

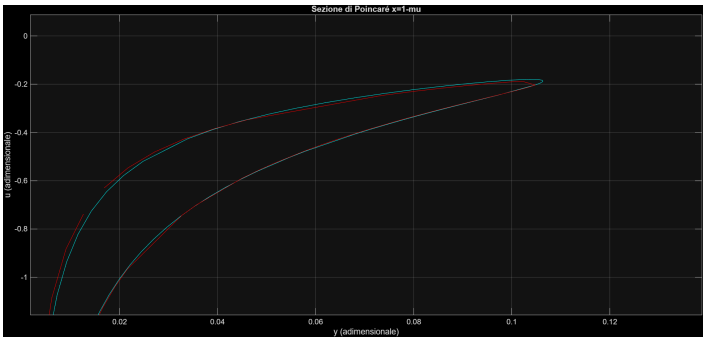


Figure 9: Poincaré Map (y vs u)

It's also important to evaluate the second Map (y vs u), to understand the sign of the velocity along the other coordinate (in this case velocity along x so u), this is necessary because of the nature of the formula to evaluate this component.

$$u_{transit} = \pm\sqrt{2U - C_{target}} \tag{4}$$

In this specific case the sign is minus, as clearly shown in the figure above.

The brackets on the M represent the position of the spacecraft at the current moment, so the chosen state is an initial state for the transit trajectory, which is a matching point in phase space, this implies that the particle flows naturally from the vicinity of L_2 to that of L_1, effectively completing the $\{X, M, E\}$ sequence.

Variable	Value
x	0.98785
y	0.06000
u	-0.46376
v	0

Table 1: Initial State Vector X,M,E

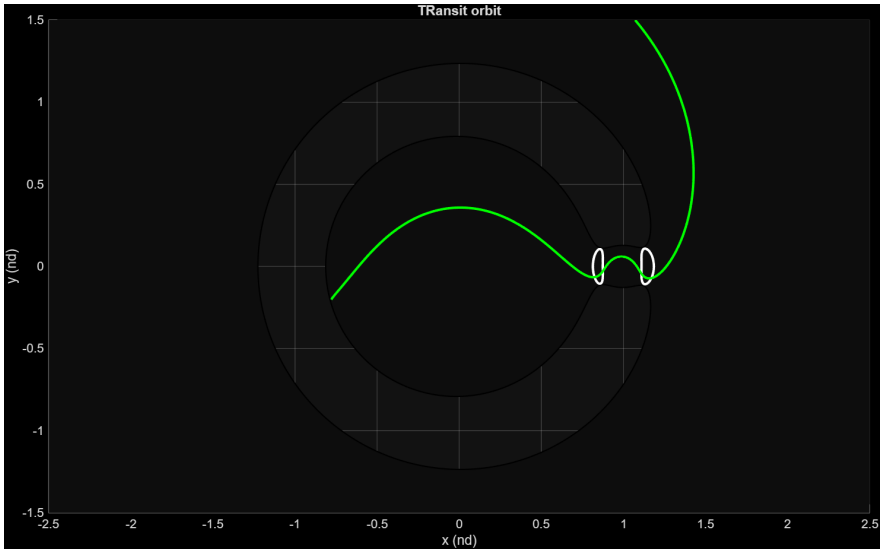


Figure 10: X,M,E transit trajectory

4.2 Transit Trajectory: $\{E, [M], X\}$

Similar to the previous point, this trajectory represents a transit, this time from the Earth realm to the external one. The concept is the same, but the manifolds traversed are different.
This time the goal is to move from the Earth realm toward the L1 point, then to the Moon realms and finally into the outer space, so the first manifold will be L1 stable, then L1 unstable with a connection at the Poincaré section with L2 stable, and, finally, L2 unstable to the X realm.

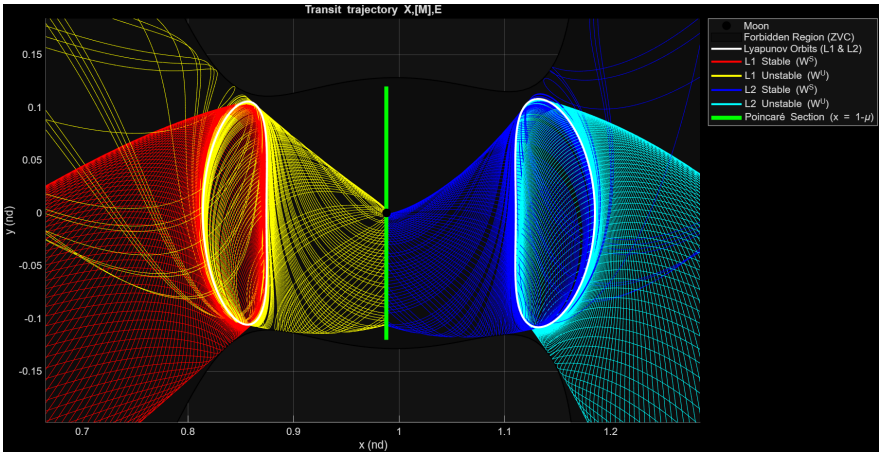


Figure 11: Transit E,M,X

This time the figure is zoomed in on the M realms, to focus the connection at the Poincaré section. From this perspective it is easier to understand where is the connection. The next step is the same as the previous trajectory, analyze the Poincaré Maps, reconstruct the initial state vector and compute the full transfer trajectory.

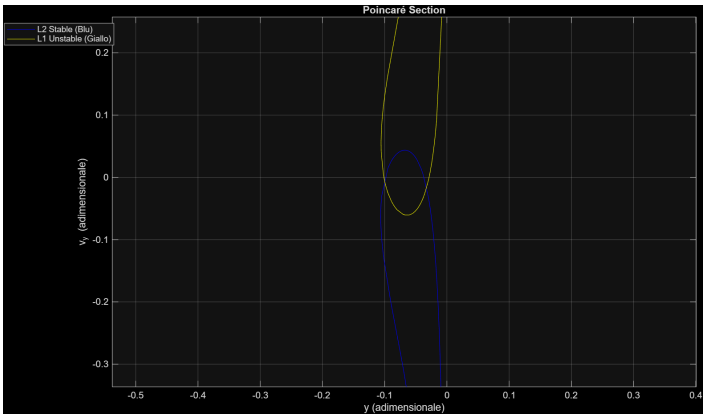


Figure 12: Poincaré Map (y vs v)

This graph is the same as the one for the previous trajectory, but with a different sensitivity on the x-axis, to have a clear vision of the intersection area.

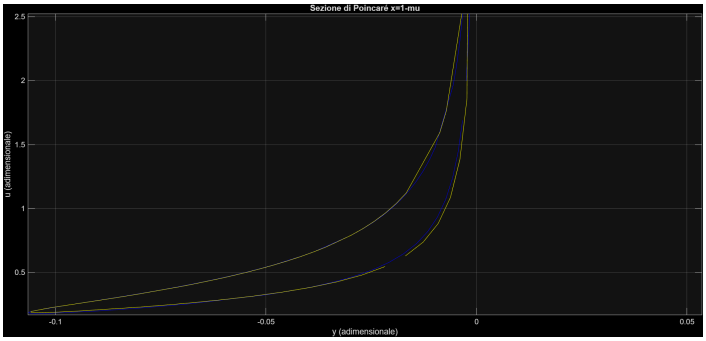


Figure 13: Poincaré Map (y vs u)

From this image is clear that this time the value of u is positive. This leads to the following state vector.

Variable	**Value**
x	0.98785
y	-0.06000
u	0.46376
v	0

Table 2: Initial State Vector E,M,X

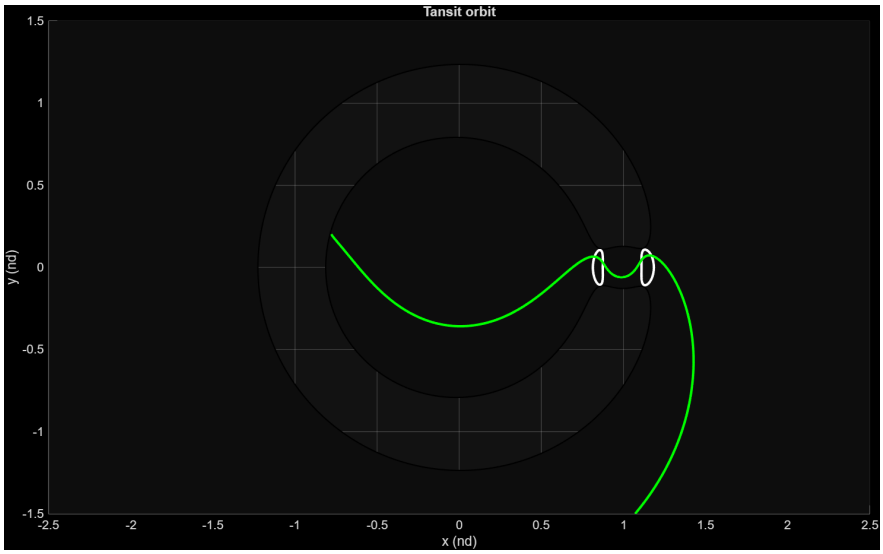


Figure 14: E,M,X transit trajectory

4.3 Non-Transit Trajectory: $\{X, [M], X\}$

This itinerary describes a temporary capture scenario; the spacecraft enters the Moon realm from the Exterior, performs one revolution around the Moon, and returns to the Exterior realm without crossing into the Earth realm.

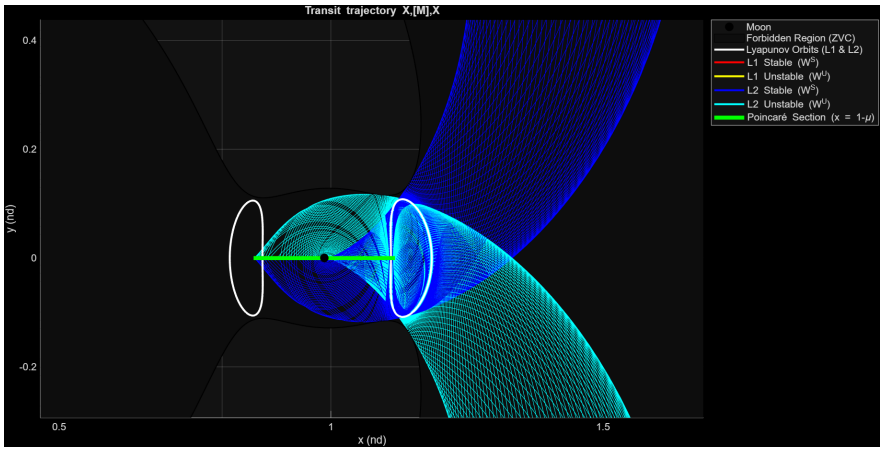


Figure 15: Non-Transit X,M,X

To evaluate this type of trajectory it is necessary to change the Poincaré section, the optimal one will be a horizontal section at y=0. This type of section can lead to calculation problems due to its proximity to the Moon. When the probe approaches the Moon too closely the gravity gradient tends to infinity, and so does the velocity, so the manifolds diverge. To avoid this problem, a condition was set to stop the propagation once the spacecraft is too close to the moon (R-safe = 0.005 normalized).

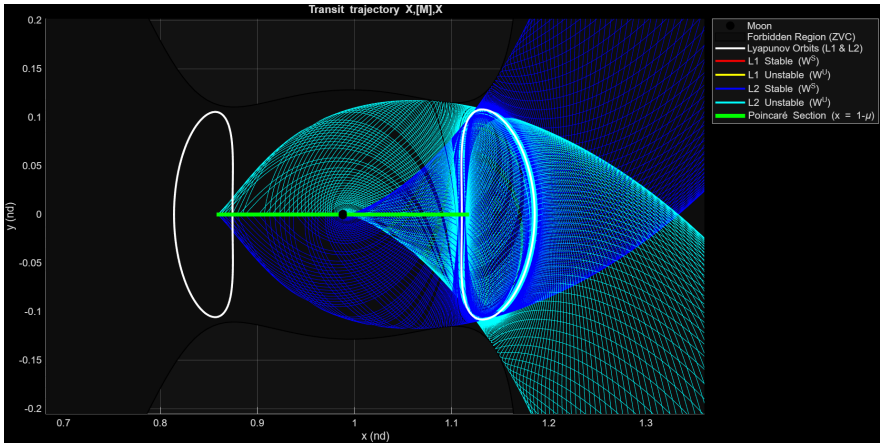


Figure 16: Non-Transit X,M,X zoomed in

The analysis of the Poincaré section is similar to the one for the transit cases, but this time the graph shows the x component of the position with respect to the velocity components.
The trajectory is composed of L2 stable manifold from X to M, to approach L2, the

L2 unstable from L2 to the Moon, then the crossing point between L2 unstable and L2 stable until it arrives again in proximity of L2 then L2 unstable to move towards the X realm.
The common area allows the transfer from L2 unstable to L2 stable and the sign of v is determined by the x vs v graph.

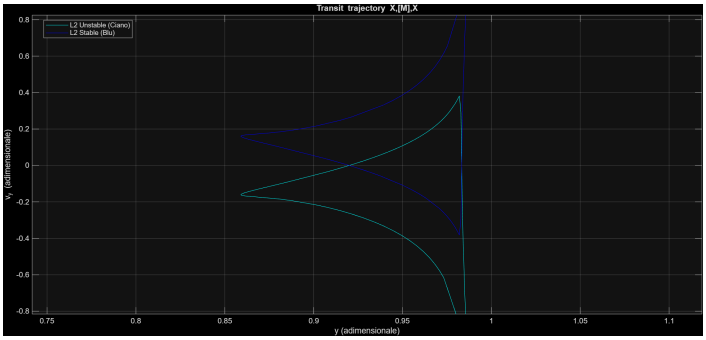


Figure 17: Poincaré Map (x vs u)

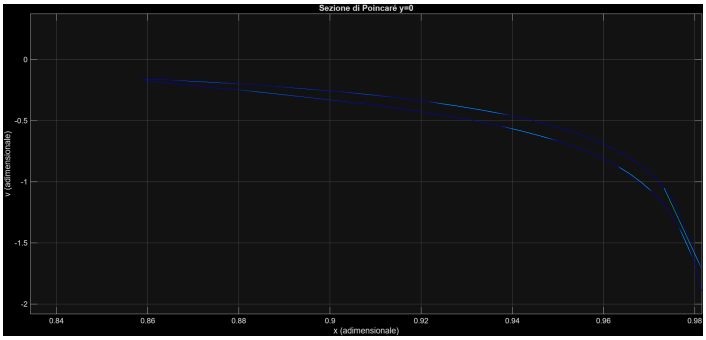


Figure 18: Poincaré Map (x vs v)

The initial state now belongs to the y=0 section, after an analysis of the graphs, the chosen state is:

Variable	**Value**
x	0.96000
y	0
u	0
v	-0.82753

Table 3: Initial State Vector X,M,X

The velocity along y is zero, because of the simmetry of this case, the goal is to cross the Poincaré section perpendicularly to enter the other L2 manifold.

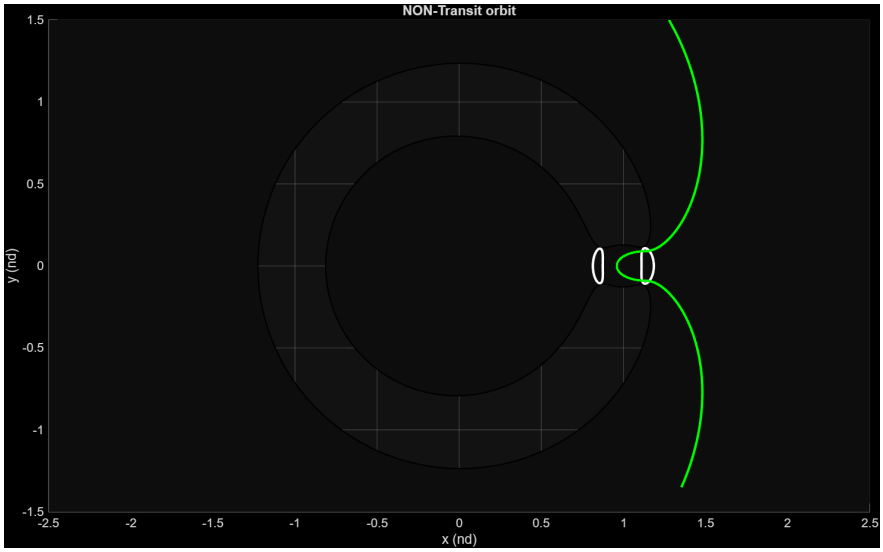


Figure 19: X,M,X non transit trajectory

4.4 Non-Transit Trajectory: $\{E, [M], E\}$

This itinerary describes a return trajectory. The spacecraft starts from the Earth realm, enters the Moon region, and comes back towards Earth without accessing the Exterior realm.

Analogous to the previous case, the intersection is identified on the horizontal Poincaré section, but this time the manifolds belong to the L1 family. It starts with a L1 stable toward the Lagrange point, then L1 unstable around the Moon, then the connection on the y=0 section with the L1 stable to go back toward the Lagrange point, and, finally, L1 unstable into the Earth realm.

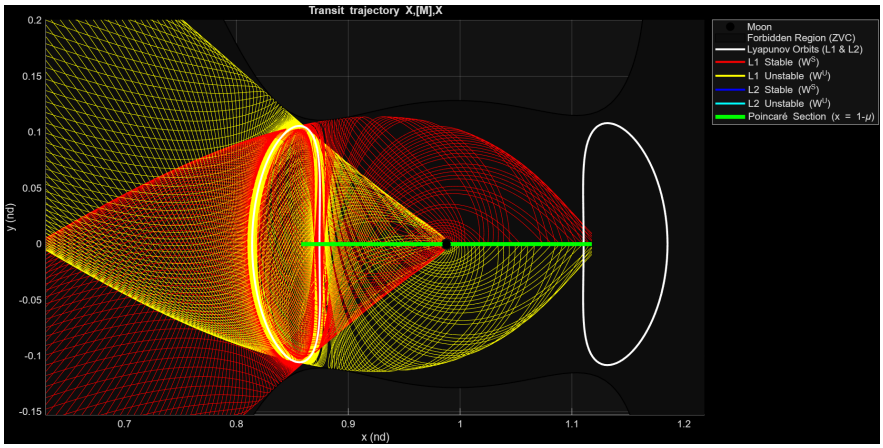


Figure 20: Non-Transit E,M,E

The existence of this connection demonstrates the "blocking" effect of the manifolds, which prevents transport between the Earth and Exterior realms for specific energy levels and initial conditions.

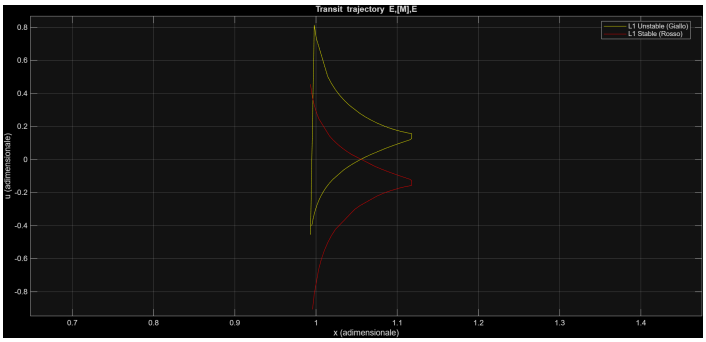


Figure 21: Poincaré Map (x vs u)

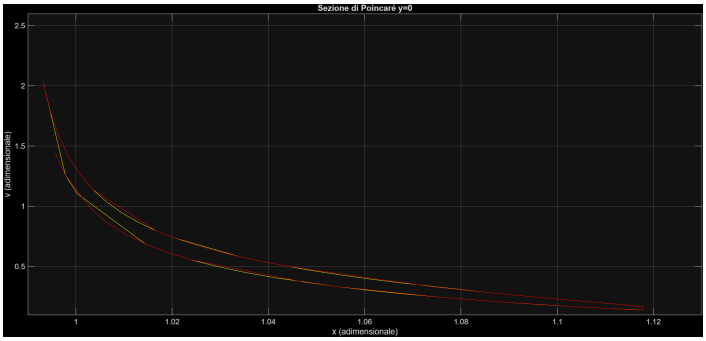


Figure 22: Poincaré Map (x vs v)

Variable	**Value**
x	1.02000
y	0
u	0
v	0.75423

Table 4: Initial State Vector E,M,E

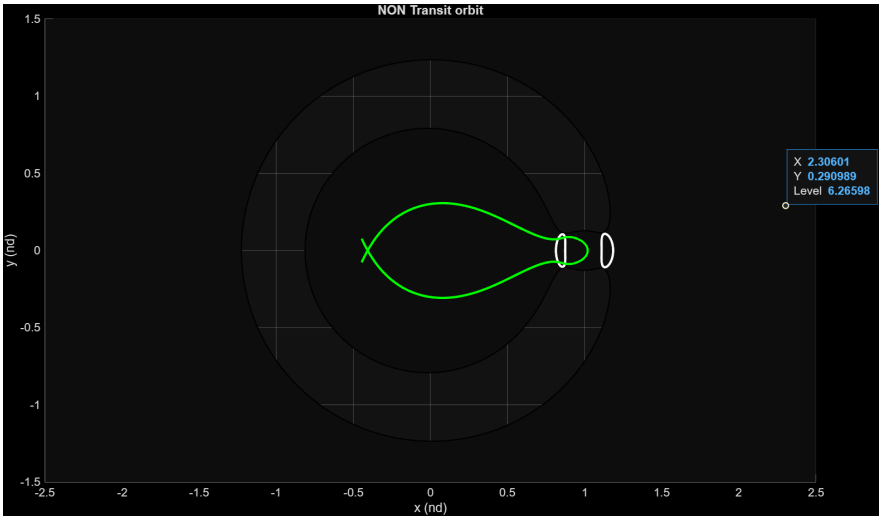


Figure 23: E,M,E non transit trajectory